chewR Quick Start Guide

A short tutorial for using the chewR package

1. What chewR does

chewR is an R package for processing scanned gum images and summarizing chewing efficiency using hue-based image analysis. The main function is run_chewr_pipeline().

2. Accepted image formats

chewR accepts the following file types: .png, .jpg, .jpeg, .tif, and .tiff.

3. Before you start

- Scan all gum images clearly and save them as image files (.png if possible)
- Put all images for analysis into a single folder
- Make the folder with all the images your working directory in RStudio
- Name the files using one of the accepted filename conventions below.

4. Filename conventions

Chew protocol

Use this when the filename contains the number of chews.

- 1_5chews_side1.png
- 1_5chews_side2.png
- 1_10chews_side1.png
- 1_10chews_side2.png
- 1_20chews_side1.png
- 1_20chews_side2.png
- 2_20chews_side1.png

Rules: ID must be numeric only; side must be side1 or side2; chew count must be numeric.

  - Here, 1 refers to the participant ID, the next string of text after the underscore refers to the number of chews, the final string of text after the second underscore refers to the front or back of the scan (side 1 and side 2)

Fixed protocol

Use this when the filename does not contain the number of chews.

- 1_side1.png
- 1_side2.png
- 2_side1.png
- 2_side2.png

Rules: ID must be numeric only; side must be side1 or side2; chew count is automatically assigned as 20.

- Here, 1 refers to the participant ID, the final string of text after the underscore refers to the front or back of the scan (side 1 and side 2)

Rejected filename examples

- PSU12_20chews_side1.jpg
- ID_3_5chews_side2.png
- PSU12_side1.jpg

5. Set your working directory in R

In R or RStudio, set your working directory to the folder containing your images.

1. Click **Session** in the top menu.
2. Click **Set Working Directory**.
3. Click **Choose Directory**.
4. Select the folder containing your gum images.

RStudio will automatically run a command like:

```
setwd("C:/path/to/your/image_folder")
```

You can confirm your working directory by running:

```
getwd()
```

6. Install, load the package, and run the pipeline

Run the following code in your R:

```
install.packages("devtools")

devtools::install_github("JohnLongPhD/chewR")
```

If devtools does not install from the code above, you may need to manually install them at: https://cran.r-project.org/web/packages/devtools/index.html

If your filenames use the chew-count format:

```
library(chewR)

pipeline_results <- run_chewr_pipeline(
  protocol           = "chew",
  output_excel_path  = "chewR_results.xlsx",
  output_plot_dir    = "chewR_plots",
  auto_crop          = TRUE,
  save_analysis_images = TRUE
)

pipeline_results
```

If your filenames use the fixed format:

```
library(chewR)

pipeline_results <- run_chewr_pipeline(
  protocol           = "fixed",
  output_excel_path  = "chewR_results.xlsx",
  output_plot_dir    = "chewR_plots",
  auto_crop          = TRUE,
  save_analysis_images = TRUE
)
```

In both formats, run the run_chewr_pipeline and these are the various modifications:

- protocol = “chew” or “fixed”
  - Chew = you had participants chew different number of times
  - Fixed = you had participants chew one sample of gum
- output_excel_path = the name of your excel spreadsheet that is generated once all your images have been analyzed
- output_plot_dir = the name of the folder that will contain the images used in the analysis
- auto_crop = “TRUE” or “FALSE”
  - Run true if you want your images to be auto cropped for analysis and if you have not manually cropped your images
  - Run false if your images have already been cropped
- save_analysis_images = “TRUE” or “FALSE”
  - Run true if you want your images to be saved that were used for analysis
  - Run false if you don’t want the images that were used for analysis to be saved

7. What the pipeline creates

- An Excel file, usually named chewR_results.xlsx
- An output folder, usually named chewR_plots
- A subfolder of cropped analysis images if save_analysis_images = TRUE

8. What is in the Excel file

The Excel workbook contains two sheets:

- summary_sides_combined: pooled result across both sides
- summary_by_side: side-specific result

Use the summary_sides_combined data for analyses.

You should also see score columns including chewr_raw_score and chewr_chewing_efficiency_score

- chewr_raw_score is the **raw** variable based on the H_SD where lower values indicate better chewing and higher values indicate worse chewing
- chewr_chewing_efficiency_score is the **transformed** variable where it is linearly transformed so that lower values indicate worse chewing and higher values indicate better chewing
- **We suggest using the transformed variable when reporting data in abstracts, conferences, manuscripts, etc.**

9. How to cite chewR

If you use chewR in research, please cite:

Long JW (2026). chewR: An Open-Source R Package for Quantifying Chewing Efficiency.

R package version 0.0.1.

https://github.com/JohnLongPhD/chewR

You can also retrieve the citation directly in R with:

```
citation("chewR")
```